

# Joshua SAKTHIVEL RAJU

Developer ~ Engineer

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🔗 joshuar26.github.io/Portfolio/

## SUMMARY

Dynamic and enthusiastic professional seeking a challenging role where I can utilize my skills and expertise in software development, engineering, and project management. Eager to contribute to innovative projects, collaborate with diverse teams, and continuously grow in a dynamic work environment.

## SKILLS

**Languages:** Python, Java, JavaScript, HTML, SCSS, PHP, C, C++

**Technologies:** Deep Learning, Machine Learning, Computer Vision, Web Development, Game Development

## EDUCATION

|                |                                                                                                                       |                                          |
|----------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 2022 - Present | <b>B.TECH in Computer Science with Specialization in Artificial Intelligence &amp; Machine Learning</b><br>CGPA: 8.48 | Vellore Institute of Technology, Chennai |
| 2023 - 2022    | <b>Sr. Secondary, CBSE(XII)</b><br>Percentage: 89.2%                                                                  | Delhi Public School, Navi Mumbai         |
| 2023 - 2020    | <b>Secondary, CBSE(X)</b><br>Percentage: 97.4%                                                                        | Delhi Public School, Navi Mumbai         |

## PROJECTS

|                                  |                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deep Learning<br>Computer Vision | <b>Ocular Disease Intelligent Recognition</b> <ul style="list-style-type: none"><li>Project Lead</li><li>To assist medical professionals and institutions in the field of ophthalmology, detect ocular diseases with remarkable precision and accuracy.</li><li><b>Tech Stack:</b> <i>Tensorflow, OpenCV</i></li></ul>                                       |
| Deep Learning<br>Computer Vision | <b>Surgical Tool Segmentation</b> <ul style="list-style-type: none"><li>Applied U-Net for precise surgical tool segmentation in real-time video footage, enhancing surgical navigation and efficiency.</li><li><b>Tech Stack:</b> <i>PyTorch, Tensorflow, OpenCV</i></li></ul>                                                                               |
| Web Application                  | <b>sMile</b> <ul style="list-style-type: none"><li>Winner of ACM (Association for Computing Machinery) WebDex 2024 Hackathon</li><li>Team Lead</li><li>A dynamic hub to connect local volunteers with community organizations, facilitating impactful engagement and collaboration.</li><li><b>Tech Stack:</b> <i>HTML, SCSS, JavaScript, PHP</i></li></ul>  |
| Game Application                 | <b>Cloud's Rescue</b> <ul style="list-style-type: none"><li>Lead Art Director</li><li>Developed characters, environment designs, level designs, and animations.</li><li>An educational 2D mobile game designed to reinforce mathematical concepts through an engaging game experience.</li><li><b>Tech Stack:</b> <i>Unity, Blender, Photoshop</i></li></ul> |

## WORK EXPERIENCE

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2/2024 - Present | <b>Undergraduate Researcher, Machine Learning</b><br>University of Pretoria, South Africa <ul style="list-style-type: none"><li>Team lead for Augmentation Team at collaboration with University of Pretoria's Data Science for Social Good Lab under Prof. Vukosi Marivate.</li><li>Worked on Text augmentation for multilingual NLP models (African languages) with 150,000+ users in the base repository.</li></ul> |
| 5/2023 - 9/2023  | <b>Internship on Game Design</b><br>Geons Logix Pvt. Ltd. <ul style="list-style-type: none"><li>Developed the game 'Cloud's Rescue'.</li></ul>                                                                                                                                                                                                                                                                         |

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|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 4/2024 – Present Promoted | <b>Vice President</b><br>• Organized events and leading initiatives to promote data science awareness and community engagement.                                                                                                              | <b>Data Science Club</b>                 |
| 7/2023 – Present Promoted | <b>Editor Head</b><br>• Editor Head of VIT, Chennai's official Newsletter- SCOPEBytes, under School of Computer Science and Engineering's Dean Sir.<br>• Acted as the bridge between the Content and Design team, and the Dean Sir's Office. | <b>SCOPEBytes Newsletter</b>             |
| 2/2023 – Present          | <b>Data Scientist</b><br>• Served as a technical member, actively contributing to project collaboration and technical support provision.                                                                                                     | <b>Google Developer Student Club</b>     |
| 3/2024 – Present          | <b>Jr. Core Committee</b><br>• Volunteered to conduct events with the aim of providing quality education for under-privileged students.                                                                                                      | <b>VITeach (Non-Profit Organization)</b> |

**PUBLICATIONS**

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | Krishna, K.V Hitesh, Joshua S Raju, and Kanimozhi S. <i>"TamilHRNet: Three-stage segmentation network for Tamil handwritten document digitalization."</i> (2024) (in press) |
| 2024 | Riddhiman Raguraman, Joshua Sakthivel Raju, and Subbulakshimi P. <i>"Adaptive NPC in Serious Games Using Artificial Intelligence"</i> (2024) (in press)                     |

**ACHIEVEMENTS**

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| 22/3/2024 | <b>ACM WebDex 2024 Hackathon</b><br>• Won the First Place in the ACM (Association for Computing Machinery) WebDex 2024 Hackathon<br>• Team Lead                                         | <b>Winner</b> |
| 27/9/2023 | <b>Game-A-Thon Hackathon</b><br>• Won the First Place in the Game-A-Thon organized by TechnoVIT'23 under the School of Computer Science and Engineering, VIT, Chennai.<br>• Lead Artist | <b>Winner</b> |

**SPOKEN LANGUAGES**

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**Fluent:** English, Hindi  
**Basic:** Telugu, Tamil, Spanish